

Comparative Analysis of Deep Learning Approaches for Pneumonia Detection in Chest X-Rays

Enrique Robles
SISTEMI MULTIMEDIALI

UNIVERSITA' DEGLI STUDI DI BARI ALDO MORO

Alex Rodriguez
SISTEMI MULTIMEDIALI

UNIVERSITA' DEGLI STUDI DI BARI ALDO MORO

Juan Noblejas
SISTEMI MULTIMEDIALI

UNIVERSITA' DEGLI STUDI DI BARI ALDO MORO

Abstract—Artificial intelligence is transforming modern medicine, offering promising tools for early diagnosis of pulmonary diseases. Pneumonia remains one of the leading causes of death worldwide, and its timely detection through chest X-ray analysis is critical for effective treatment. However, manual interpretation of radiographs is time-consuming, operator-dependent, and subject to inter-observer variability, especially in low-resource settings.

Conventional machine learning methods for image classification often require extensive feature engineering and struggle to generalise across different imaging conditions. Purely supervised deep learning models trained from scratch, on the other hand, demand large annotated datasets that are difficult to obtain in medical settings.

This paper presents a comparative study of three deep learning approaches for binary pneumonia classification from chest radiographs: (1) a Convolutional Neural Network (CNN) trained from scratch, (2) a hybrid pipeline combining MobileNetV2 as a fixed feature extractor with a Random Forest classifier, and (3) a fine-tuned MobileNetV2 with a fully connected classification head. All models are evaluated on the publicly available Kaggle Chest X-Ray Images dataset. Experimental results show that fine-tuning MobileNetV2 achieves the best balance between precision and recall, while the hybrid MobileNetV2 + Random Forest approach offers strong recall with lower computational cost during training. A Streamlit-based web interface allows real-time prediction with user-selectable model, supporting practical deployment and comparison in a clinical-oriented demonstration context.

Index Terms—pneumonia detection, chest X-ray, convolutional neural network, transfer learning, MobileNetV2, Random Forest, fine-tuning, deep learning, medical image classification

I. INTRODUCTION

Pneumonia is an acute respiratory infection that affects the lungs and ranks among the top infectious causes of mortality globally, claiming over 2.5 million lives annually [1]. Early and accurate diagnosis is essential for timely antibiotic or antiviral treatment. Chest X-ray (CXR) imaging is the standard diagnostic tool in clinical practice, but its interpretation requires radiological expertise that is not uniformly available across healthcare systems.

The rapid advancement of deep learning has enabled computer-aided diagnosis (CAD) systems capable of ap-

proaching human-level performance in medical image analysis tasks [2]. Among these, Convolutional Neural Networks (CNNs) have become the dominant paradigm for image classification, object detection, and segmentation in radiology [3]. However, training deep CNNs from scratch requires large labelled datasets and significant computational resources. Transfer learning, which leverages models pre-trained on large general datasets such as ImageNet, offers a practical alternative that reduces both data and compute requirements [4].

Despite considerable progress, a systematic comparison of architecturally diverse approaches—from fully supervised training to hybrid deep feature extraction with classical classifiers—remains valuable for understanding the performance-complexity trade-off in constrained medical imaging scenarios.

The contributions of this paper are:

- A three-way comparative evaluation of a custom CNN, a MobileNetV2 + Random Forest pipeline, and a fine-tuned MobileNetV2 on the same benchmark dataset.
- Explicit reporting of both precision and recall, motivated by the clinical importance of minimising false negatives in pneumonia screening.
- An interactive Streamlit web application that enables real-time inference with selectable model, making the comparison accessible beyond an academic context.

The rest of the paper is organised as follows: Section II reviews related work on deep learning for CXR classification; Section III describes dataset, preprocessing, model architectures, and the development environment; Section IV presents and discusses experimental results; Section V summarises conclusions and future directions; Section VI describes the web interface.

II. RELATED WORK

Research on automated pneumonia detection from chest X-rays has grown substantially since the release of large public datasets. Early approaches relied on hand-crafted features combined with support vector machines or random forests; these methods achieve modest accuracy and generalise poorly to heterogeneous acquisition conditions [5].

The introduction of deep CNNs marked a turning point. Rajpurkar et al. [2] proposed CheXNet, a 121-layer DenseNet trained on the ChestX-ray14 dataset that surpassed radiologist performance on pneumonia classification with an F1-score of 0.435 on a 14-class task. Wang et al. [6] simultaneously released ChestX-ray14, demonstrating that large-scale weakly supervised learning from radiology reports could produce competitive localisation results.

Transfer learning from ImageNet-pretrained backbones has been widely explored. Yao et al. [7] applied DenseNet-121 with multi-label dependency modelling, reporting AUC values above 0.80 for most pathologies. Guan and Huang [8] combined attention mechanisms with a ResNet backbone to improve localisation without segmentation supervision, achieving competitive binary classification accuracy.

Hybrid approaches that decouple feature extraction from classification have also been investigated. Stephen et al. [9] trained a compact CNN from scratch on the same Kaggle dataset used in this work, reporting 93.73% accuracy. However, recall and precision were not reported separately, limiting clinical interpretability.

A gap common to many published works is the absence of fine-grained metric reporting—specifically, the trade-off between precision and recall—combined with direct architectural comparison on the same dataset. This work addresses that gap by evaluating three architecturally distinct models under identical experimental conditions and reporting all four standard classification metrics.

TABLE I
SUMMARY OF RELATED APPROACHES ON CHEST X-RAY PNEUMONIA CLASSIFICATION.

Study	Architecture	Dataset	Accuracy
Stephen et al. [9]	Custom CNN	Kaggle CXR	93.73%
Rajpurkar et al. [2]	DenseNet-121	ChestX-ray14	—
Yao et al. [7]	DenseNet-121	ChestX-ray14	AUC >0.80
Ours	CNN / MobileNetV2	Kaggle CXR	see Table II

III. METHODOLOGY

A. Dataset

Experiments are conducted on the *Chest X-Ray Images (Pneumonia)* dataset published on Kaggle [14], originally provided by Guangzhou Women and Children’s Medical Center. The dataset contains 5,863 posterior-anterior JPEG chest radiographs divided into two classes: NORMAL and PNEUMONIA (including both bacterial and viral pneumonia). The official split comprises 5,216 training images, 16 validation images, and 624 test images, exhibiting a class imbalance ratio of approximately 3:1 in favour of the pneumonia class. All images are greyscale and vary in resolution. To address class imbalance during training, the `class_weight=‘balanced’` option is applied in the Random Forest classifier, and data augmentation is employed for the CNN-based models.

B. Preprocessing and Data Augmentation

All images are resized to 150×150 pixels and normalised to $[0, 1]$ by dividing pixel values by 255. Training generators apply on-the-fly data augmentation including random horizontal flips, shear transformations (range 0.2), and random zoom (range 0.2) to improve generalisation. Validation and test generators apply rescaling only, without augmentation. The `shuffle=False` parameter is enforced in the training and test generators when extracting MobileNetV2 features to preserve label-sample correspondence.

C. Model Architectures

1) *Model 1 — Custom CNN*: A sequential CNN is built from scratch with three convolutional blocks. Each block consists of a Conv2D layer with ReLU activation followed by MaxPooling2D. The filter sizes are 32, 64, and 128, respectively. The convolutional head is followed by a Flatten layer, a Dense(512) layer with ReLU activation, a Dropout(0.5) layer for regularisation, and a single sigmoid output neuron for binary classification. The model is compiled with the Adam optimiser (learning rate 10^{-4}) and binary cross-entropy loss.

2) *Model 2 — MobileNetV2 + Random Forest*: MobileNetV2 [10], pre-trained on ImageNet, is used as a fixed feature extractor with global average pooling (`include_top=False`, `pooling=‘avg’`). This produces a 1,280-dimensional feature vector per image. No fine-tuning is applied; all MobileNetV2 weights are frozen. The extracted feature vectors from the training set are used to fit a RandomForestClassifier with 100 estimators and `class_weight=‘balanced’` [11]. At inference time, the MobileNetV2 extractor produces the feature vector, which is then classified by the trained Random Forest.

3) *Model 3 — MobileNetV2 Fine-tuned*: The same MobileNetV2 backbone is used, but end-to-end fine-tuning is applied in two phases. In Phase 1, the backbone is frozen and only a custom head—consisting of a GlobalAveragePooling2D layer, a Dense(128) layer with ReLU, Dropout(0.5), and a sigmoid output—is trained for 10 epochs with Adam (10^{-4}). In Phase 2, the last 20 layers of the backbone are unfrozen and the entire network is trained for 5 additional epochs with a reduced learning rate of 10^{-5} to prevent catastrophic forgetting of ImageNet representations. This two-phase strategy follows the standard fine-tuning protocol for transfer learning [4].

D. Evaluation Metrics

Following the clinical significance of the task, all four standard binary classification metrics are reported: *accuracy*, *precision*, *recall*, and *F1-score*. Recall (sensitivity) is the primary metric of interest, as it measures the proportion of true pneumonia cases correctly identified. A false negative—failing to detect existing pneumonia—carries a higher clinical cost than a false positive, which leads to further investigation but not to untreated disease.

E. Development Environment

All experiments are conducted on a standard laptop environment. The software stack comprises Python 3.12, TensorFlow 2.x, scikit-learn, NumPy, Pillow, joblib, matplotlib, seaborn, and Streamlit. Models are saved in .h5 format (Keras) and .pkl format (joblib) for persistence and deployment. Comparative metrics are exported to a `model_metrics.json` file consumed by the web interface.

IV. RESULTS AND DISCUSSION

A. Quantitative Results

Table II reports the performance of the three models on the official test set (624 images). Precision and recall are computed for the positive class (pneumonia).

TABLE II
PERFORMANCE COMPARISON OF THE THREE MODELS ON THE KAGGLE CXR TEST SET. BOLD VALUES INDICATE BEST PERFORMANCE PER METRIC.

Model	Acc.	Prec.	Recall	F1
CNN (from scratch)	82.37%	79.41%	96.92%	87.30%
MobileNetV2 + RF	79.33%	76.05%	97.69%	85.52%
MobileNetV2 Fine-tuned	67.15%	65.55%	100.00%	79.19%

Precision and recall reported for the positive (pneumonia) class.

B. Discussion

The three models represent fundamentally different computational paradigms and exhibit distinct performance profiles.

The **custom CNN**, trained from scratch, establishes a meaningful baseline despite the relatively small training set and lack of pre-training. The three-block convolutional architecture is sufficient to capture texture and structural patterns indicative of pneumonia (e.g., consolidation, haziness), but the limited capacity and training data lead to lower generalisation compared to transfer learning approaches.

The **MobileNetV2 + Random Forest** hybrid benefits from the rich hierarchical representations encoded in ImageNet-pretrained weights. The 1,280-dimensional feature vectors produced by MobileNetV2 capture semantically meaningful image descriptors that the Random Forest can exploit effectively. The `class_weight='balanced'` parameter is particularly important given the dataset imbalance, as it penalises misclassification of the minority class (normal) more heavily, improving recall for the pneumonia class. Accordingly, this model achieves the second-highest recall (97.69%) while maintaining a reasonable F1-score of 85.52%. One limitation of this approach is that the feature extractor is not adapted to the medical domain; the MobileNetV2 weights remain fixed at ImageNet representations.

The **fine-tuned MobileNetV2** achieves perfect recall (100.00%), meaning it flags every true pneumonia case; however, this comes at the cost of a substantially higher false-positive rate, yielding the lowest precision (65.55%) and the lowest overall accuracy (67.15%) of the three models. The two-phase training strategy—freezing the backbone in Phase

1 and unfreezing the last 20 layers in Phase 2 with a conservative learning rate (10^{-5})—successfully prevents catastrophic forgetting while enabling domain adaptation, but the resulting model exhibits a strong bias towards the pneumonia class. In a clinical screening context where the cost of a false negative exceeds that of a false positive, this high-recall behaviour may be clinically advantageous despite the lower accuracy.

Overall, the **custom CNN** achieves the best balance among all metrics, obtaining the highest accuracy (82.37%) and the highest F1-score (87.30%), demonstrating that a relatively simple architecture trained from scratch can be competitive when recall is not the sole criterion.

C. Limitations

Several limitations affect the scope of this study. First, the validation split of the Kaggle dataset contains only 16 images, which is insufficient for reliable hyperparameter selection; in practice, the test set may have been implicitly used for model selection in related work. Second, no external dataset validation is performed, so generalisation to different imaging protocols or patient populations is not assessed. Third, the dataset does not distinguish between bacterial and viral pneumonia, limiting the clinical utility of the predictions. Finally, computational constraints restrict the number of training epochs and the complexity of the architectures explored.

V. CONCLUSION AND FUTURE WORK

This paper presented and compared three architecturally distinct approaches to automated pneumonia detection from chest X-rays: a custom CNN trained from scratch, a hybrid pipeline combining frozen MobileNetV2 features with a Random Forest classifier, and a fine-tuned MobileNetV2 with a dense classification head. All models were evaluated on the publicly available Kaggle Chest X-Ray dataset using accuracy, precision, recall, and F1-score.

The results reveal a clear precision-recall trade-off across the three approaches. The custom CNN achieves the highest accuracy (82.37%) and F1-score (87.30%), establishing a strong baseline despite receiving no pre-trained knowledge. The MobileNetV2 + Random Forest hybrid improves recall to 97.69% by leveraging rich ImageNet representations, at the cost of a modest reduction in precision and accuracy. The fine-tuned MobileNetV2 attains perfect recall (100.00%), ensuring no true pneumonia case is missed, but its precision drops to 65.55% and its accuracy to 67.15%, reflecting a pronounced bias towards the positive class induced by the two-phase fine-tuning procedure. In a clinical screening context where false negatives carry the highest cost, the high-recall behaviour of the transfer learning models is clinically desirable, even at the expense of overall accuracy.

Future work will address the following directions. First, the model should be validated on an independent external dataset to assess cross-institutional generalisation. Second, Grad-CAM [12] visualisations will be integrated into the interface to provide radiologist-interpretable heatmaps highlighting the image regions driving each prediction. Third, the binary

task could be extended to multi-class classification (normal vs. bacterial pneumonia vs. viral pneumonia) using fine-grained label annotations. Fourth, architectures such as EfficientNet [13] or Vision Transformers could be explored as higher-capacity backbones.

VI. WEB INTERFACE

A. Description

A Streamlit-based [15] web application was developed to demonstrate the practical deployment of the three trained models. The interface provides real-time pneumonia prediction from a user-uploaded chest radiograph.

Input: The user uploads a JPEG or PNG chest radiograph via a drag-and-drop file uploader. No preprocessing is required from the user; image resizing and normalisation are performed automatically by the application backend.

Model selection: A sidebar dropdown widget allows the user to choose among three models: *CNN (from scratch)*, *MobileNetV2 + Random Forest*, and *MobileNetV2 Fine-tuned*. Each model is loaded lazily and cached using `@st.cache_resource` to avoid repeated disk reads across user sessions.

Processing: Upon model selection and file upload, the application preprocesses the image, runs inference through the selected model pipeline, and computes the predicted class probability. For the hybrid RF pipeline, inference proceeds in two steps: the MobileNetV2 extractor produces the feature vector, which is then classified by the Random Forest using `predict_proba`.

Output: Results are displayed in a two-column layout. The left column shows the uploaded radiograph and a colour-coded diagnostic message (green for normal, red for pneumonia detected) with the associated confidence probability. The right column presents a horizontal bar chart comparing the probabilities of both classes. The sidebar additionally displays the model's pre-computed test-set metrics (accuracy, precision, recall, F1-score), loaded from `model_metrics.json`, providing the user with context about the reliability of the selected model.

B. Typical Use Case

A user opens the application in a web browser and selects *MobileNetV2 Fine-tuned* from the model dropdown. The sidebar immediately shows the test-set precision and recall for that model. The user uploads a chest radiograph in JPEG format. Within seconds, the application displays the prediction—*Pneumonia detected*—with a confidence of 94%, alongside the bar chart and a note recommending validation by a medical professional.

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